Enhancement Two Narrative

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1. Artifact Description

   The artifact I selected is the Travlr Getaways full-stack web application that I originally created in CS 465. The project includes a customer-facing travel website, an Angular administrative application, a Node.js and Express backend, REST API, and a MongoDB database. The application stores trip information and allows trips to be viewed, added, and edited. For this enhancement, I focused on improving how trip information is organized and retrieved by the backend using the trip code.

2. Justification

   Originally, finding an individual trip required the application to send a query to MongoDB using the trip code. For the enhancement, I added a JavaScript map that uses each trip code as a key and the trip as its value. When the list of trips is retrieved from MongoDB, the application builds the map by going through the trips and sorting each one using the code.

   Once the map has been created, the application can use a trip code to directly retrieve the corresponding trip instead of going through the entire collection in memory. I also updated the add and update functions so the map stays current whenever a trip is added or edited. The main tradeoff with this is that the application uses more memory to maintain the map, but it provides a more efficient way to perform repeated lookups.

3. Course Outcomes

   This enhancement mostly shows course outcome 3 and some 4. I improved the way trip information is organized by using a map as the data structure for sorting. I also considered the tradeoff between using additional memory and making more lookups more efficient.

4. Reflection

   Completing this enhancement helped me better understand how choosing a data structure can affect the way an application handles information. I learned how a JavaScript Map can use a key, like a trip code, to retrieve a specific value without having to search through every item stored in the Map. I also learned that there are tradeoffs when choosing a data structure.

   One challenge I encountered was making sure the Map stays consistent with the MongoDB database. When a trip is added or edited, the application also needs to add or update that trip in the Map. Another challenge occurred when I first changed the trip lookup to use the Map. The Edit Trip page stopped displaying all of the trip information

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because the Angular component expected the API response to be an array. I corrected this by returning the trip from the Map inside an array, which allowed the new data structure to work without changing the existing frontend. I tested the application by retrieving, adding, and editing trip information and also restarted the application to make sure everything continued to work correctly. Overall, this enhancement improved my understanding of data structures and design tradeoffs. I also learned how changes to one part of the full-stack can affect other parts of the system.